

12-April-2026

Agenda:

Statistics - 7

probability distributions:

Bernoulli, Binomial, Poisson, Exponential, Uniform, Normal, Z-score

→ next-class

Bernoulli Distribution:

Coin Toss $\rightarrow H, T$

$$x \rightarrow (\sin T_0)$$

Coin Toss $\rightarrow H, T$
 $x \rightarrow \text{Coin Toss}$
 $P(x) \rightarrow H \rightarrow 0 \rightarrow \text{Head}$
 $T \rightarrow 1 \rightarrow \text{Tain}$

0 does not mean failure

random variable

Binary $\rightarrow$ Bernoulli model

$\downarrow$
trial $\rightarrow$ output : 0 failure / no pr
 : 1 success / pr

→ simple possible distributions

→ A single trial with exactly two outcomes

$$\rightarrow 1 \rightarrow \text{sup}(lev)$$

— 0 → failure

→ Every coil flip - yes/no

→ Every question → pair / fair

→ It has only parameter $\rightarrow p$ (the prob of success)

Example :

You are applying for a job:

$x \rightarrow 1 \rightarrow$ got the job, prob p

$x \rightarrow 0 \rightarrow$ you don't get the job, prob $(1-p)$

probability formula:

$$P(X=x) = p^x (1-p)^{1-x}$$

$$x \in [0, 1]$$

$$x=0 \rightarrow p^0 (1-p)^{1-0} \rightarrow 1 \cdot (1-p)$$

$$\rightarrow \underline{1-p}$$

$$x=1 \rightarrow p^1 (1-p)^{1-1} \rightarrow p \cdot 1 \rightarrow \underline{p}$$

p.s: 60% customer buy shoes. $\rightarrow$ on an average 60% buy shoes
 $\downarrow$
 mean $\rightarrow 0.6$

$$x \rightarrow 1 \rightarrow p(X=1) = 0.6$$

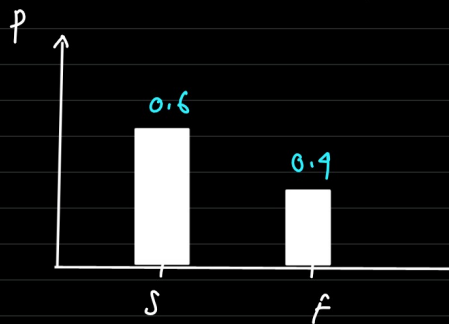
$$p \rightarrow 0.6$$

$$\rightarrow p(X=0) = 0.4 \text{ (does not buy shoe)}$$

$$\text{variance} \rightarrow p(1-p)$$

$$\rightarrow 0.6 \times 0.4$$

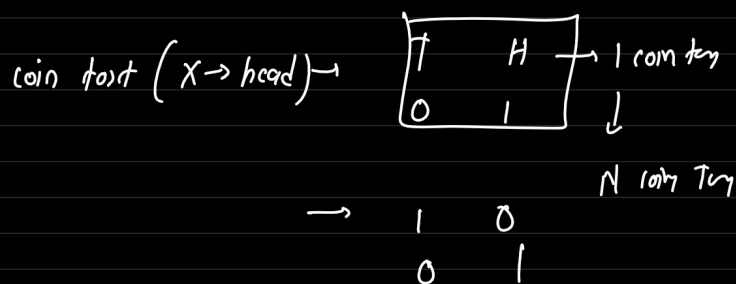
$$\rightarrow 0.24$$



Binomial distribution:

Bernoulli is discrete with only two values

: models n Bernoulli trials



coin toss, $p \rightarrow \text{Head} \rightarrow 0.7$

Toss $\rightarrow$

H	T
1	0
Toss $\rightarrow$	1
Toss $\rightarrow$	0
Toss $\rightarrow$	0
Toss $\rightarrow$	1

Ques: pizz brand $\rightarrow$ garlic dip

$\rightarrow$ Each customer buy garlic dip with a prob of 0.6

$\rightarrow p = 0.6$

$\rightarrow$ Today 5 custom arrive

$\rightarrow$ what is the prob that exactly 3 customer will buy

$\rightarrow$ combination, pmf (Bernoulli)

$$\rightarrow \binom{n}{k} \cdot p^k (1-p)^{n-k}$$

$\downarrow$
n choose k

$k \rightarrow$ exact number of success wanted $\rightarrow 3$

$n \rightarrow$ total trial $\rightarrow 5$

$p \rightarrow$ success probability $\rightarrow 0.6$

$1-p \rightarrow$ failure probability $\rightarrow 0.4$

$\binom{n}{k} \rightarrow$ number of ways to arrange success among trials.

why combination:

3 buyers out of 5

one pattern:

B	B	B	N	N
B	N	B	B	N
N	B	B	B	N

combination counts all such arrange

full calculation:

$$3! = 3 \times 2 \times 1$$

$$1! = 1 \times 1$$

step 1: combination

$$\binom{n}{k} = \frac{n!}{(n-k)!k!} = \frac{5!}{2!3!} = \frac{5 \times \cancel{4} \times \cancel{3} \times \cancel{2} \times 1}{2 \times 1 \times 3 \times \cancel{2} \times 1} = 10$$

Bernoulli $\rightarrow$ Trial - 1 $\rightarrow$ answer
 Trial - 2 $\rightarrow$ answer
 :
 Trial - 3 $\rightarrow$ answer

Binomial: Binomial(n, p) $n=10$

$$(T-1) + (T-2) + (T-3) + \dots + (T-10)$$

10 such trials

$$10 \cdot \boxed{\boxed{1}}$$

T-1, T-2, T-3 ... (T-10)
 $\rightarrow$
 Bernoulli(p)

Success part:

$$p^3 = 0.6^3 = 0.216$$

Failure part:

$$(1-p)^{5-3} = 0.4^2 \rightarrow 0.16$$

combine all parts:

$$\begin{aligned} P(X=3) &= 10 \times 0.216 \times 0.16 \\ &= 10 \times 0.03456 \\ &= 0.3456 \end{aligned}$$

Assignment:

probability of at least 3 buyers.

$$P(X \geq 3)$$

Poisson Distribution:

Models: count of rare event occurring at a constant rate (λ) in a fixed interval.

Discrete: values are non-negative integers

parameter: λ (average rate of events)

Memoryless: event occur independently

web-traffic: visit per minute
customer service: calls per hour } examples

formula:
$$p(k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!}$$

$k \rightarrow$ no of events

Example: A website average 4 visit per hour

$$\lambda = 4$$

Question: what's the prob of getting exactly 6 visit in a hour.

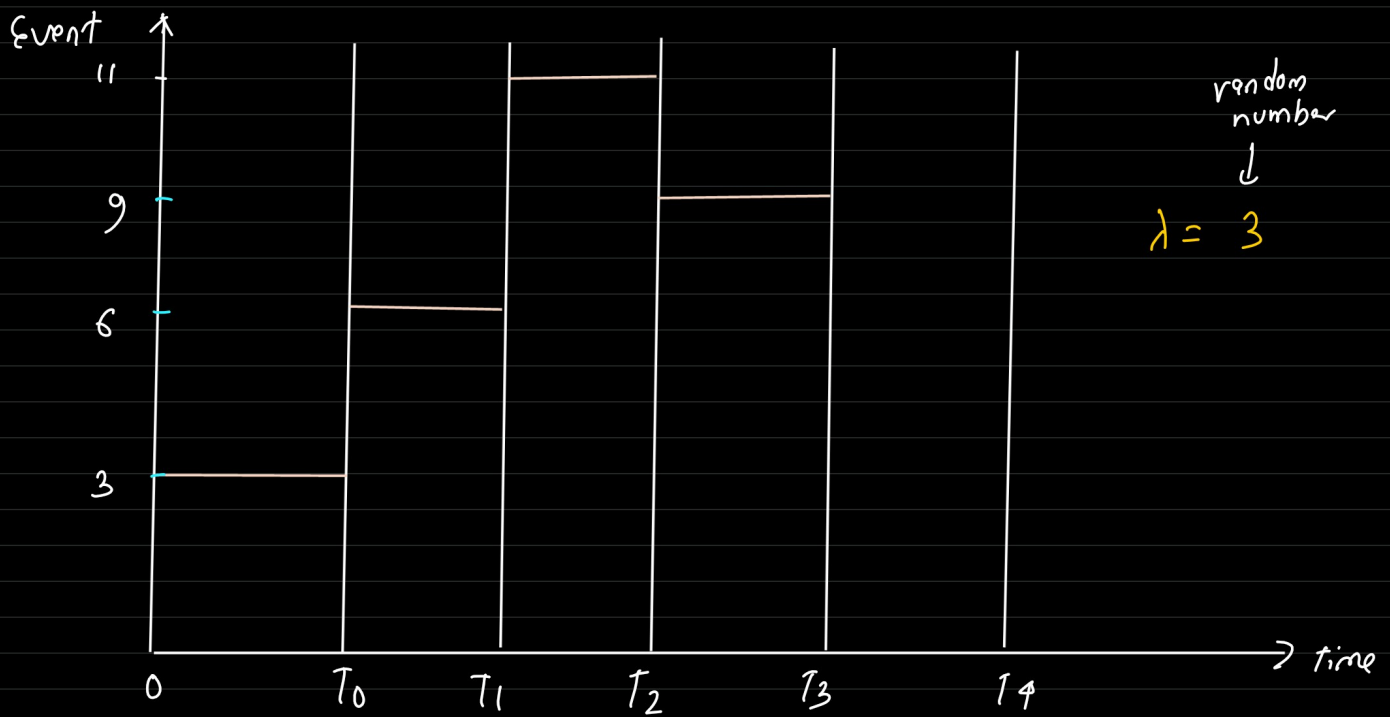
$$\begin{matrix} k=6 \\ \lambda=4 \end{matrix}$$

$$p(k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!} \approx \frac{e^{-4} \cdot 4^6}{6!}$$

$$\boxed{e = 2.7182}$$

$$\approx 0.104$$

Poisson $\rightarrow$ ML $\rightarrow$ robust



$$T_0 - 0 = 1 \text{ hr}$$

$$T_1 - T_0 = 1 \text{ hr}$$

$$T_2 - T_3 = 1 \text{ hr}$$

Exponential distribution:

Poisson: the time between events in a poisson distribution



Poisson: How many events happen in a fixed time?

Exponential: How much time until the next event.

Example.

call centre: Time between two calls

web-server: Time between two user login

server request:

on an average, 3 request per minute.

Q: what the prob that the next request comes in within 10 seconds?

Formula: $P(T \leq t) = 1 - e^{-\lambda t}$

$T \rightarrow$ waiting time until the next event

$\lambda =$ rate

$t =$ time we care about

$$\lambda (\text{poisson rate}) = 3 \text{ req/minute}$$

$$\rightarrow \text{rate per second} = \frac{3}{60} = 0.05$$

$$t = 10 \text{ (sec)}$$

$$P(T \leq 10) = 1 - e^{-0.05 \times 10} \approx 0.393$$

$\sim 39\%$ chance of getting next request within 10 seconds.

Scen: A server crashes on an average 0.1 times per day.

Q: what's the prob it runs > 10 days without crashing.

$$\lambda = 0.1$$

$$t = 10$$

$$P(T) = 1 - e^{-\lambda \cdot t}$$

$$\text{crash} = 1 - e^{-0.1 \cdot 10}$$

$$\approx 0.367 \Rightarrow 1 - 2.718^{-1}$$

$$1 - 0.3678$$

$$= 1 - 0.367 \approx \underline{\underline{0.6321}}$$

$$= \underline{\underline{0.6321}} \text{ (crash)}$$

$$(1 - P(T)) = 1 - 0.6321$$

$$= \underline{\underline{0.367}} \text{ not crash}$$

uniform distribution

Example: random arrival time between 1 PM & 2 PM

rolling dice $\rightarrow$ (1, 2, 3, 4, 5, 6)

$\rightarrow$ bus arrives everytime between 10:00 & 10:30

$\rightarrow$ 10:10 $\rightarrow$

$\rightarrow$ 10:20 $\rightarrow$

10:25

pdf

10:00

10:00

10:00

$\rightarrow$ 10:00:01

pdf $\rightarrow$



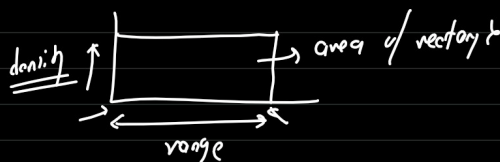
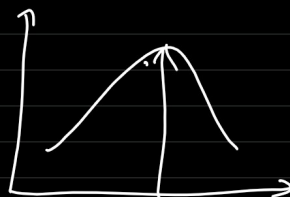




10:00

10:30





dice $\rightarrow$ (1, 2, 3, 4, 5, 6)

1 $\rightarrow$ 1 $\rightarrow$ density

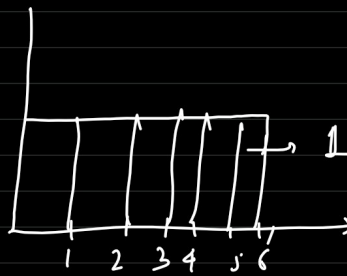
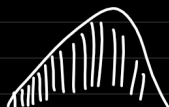
2 $\rightarrow$ 1 $\rightarrow$

3 $\rightarrow$ 1

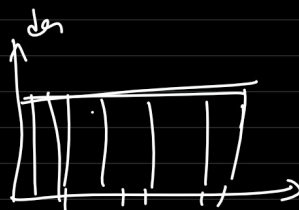
4 $\rightarrow$ 1

5 $\rightarrow$ 1

6 $\rightarrow$ 1



	minutes	ms	<u>freq</u>	density
∞ —	10:00 — 10:05 →	—	—	—
∞ —	10:05 — 10:10 →	—	—	—
∞ —	10:10 — 10:15 →	—	—	—
∞ —	⋮ →	—	—	—
∞ —	10:25 — 10:30 →	—	—	—



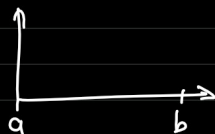
→ Every value in the interval $[a, b]$ is equally likely.

→ The pdf is a flat rectangle, constant height

$$\text{width} = b - a$$

$$\text{height} \times \text{width} = 1$$

↑
total
prob



$$\text{height} = \frac{1}{\text{width}} = \frac{1}{b-a}$$

Scenario: — A rider can arrive anytime
between 20 min — 30 min

Assume every time in that interval is equally likely

$$\text{so: } X \sim \text{Uniform}(20, 30)$$

Ques: what is the prob delivery arrives between 22 & 25 minutes?

$$a = 20$$

$$b = 30$$

$$\text{height} = \frac{1}{b-a} = 0.1$$

$$= P(22 \leq X \leq 25)$$

prob = area of rectangle

$$= \text{width} \times \text{height}$$

$$\text{width} = 3$$

$$\text{height} = 0.1$$

$$\begin{aligned} \text{so prob} &= \text{width} \times \text{height} \\ &= 3 \times 0.1 = 0.3 \end{aligned}$$

